

Notes: Kaul, Mehrotra, & Morck (JF, 2000)
Demand Curves for Stocks Do Slope Down: New Evidence from an Index
Weights Adjustment

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1 Big picture

- **Question.** Do demand curves for individual stocks slope downward, or are stocks close substitutes with nearly flat demand curves?
- **Empirical setting.** The paper studies a mechanical November 1996 Toronto Stock Exchange (TSE) 300 index-weight adjustment caused by a change in the definition of public float.
- **Policy/event shock.** The TSE changed the blockholding threshold used in the public-float definition from 15% to 20%. Stakes between 15% and 20% became part of the public float.
- **Affected stocks.** Thirty-one TSE 300 stocks had economically large float increases and therefore index-weight increases. Index funds tracking the TSE 300 had to buy these stocks.
- **Why this is clean.** The event was announced three months earlier, used public ownership data, matched the Ontario Securities Commission definition, and did not add firms to the index. Therefore it is designed to isolate demand rather than information.
- **Main event-week result.** In week 0, affected stocks earn 4.31% mean returns versus 1.97% for comparison stocks. The mean excess return is 2.34% and the beta-adjusted abnormal return is 2.07%.
- **Volume result.** Event-week median excess turnover is 1.000 for affected stocks versus 0.395 for comparison stocks, consistent with index-fund rebalancing.
- **No short-run reversal.** Trading volume returns to normal within about two weeks, but the price increase does not reverse over the following weeks.
- **Cross-sectional evidence.** Larger float changes and larger index-weight changes are associated with larger event-week abnormal returns and greater turnover.
- **Economic takeaway.** A non-informational demand shock moves prices. The most natural interpretation is that demand curves for stocks slope down.

2 Data and measurement

2.1 The TSE public-float redefinition

- The TSE 300 index was value weighted using each stock's public float.
- Before the event, the TSE public float included stakes below 15%.
- The November 15, 1996 rule change included all stakes below 20%.
- The new definition matched the OSC's 20% threshold for controlling stakes.
- The event was announced to institutional investors on August 6, 1996.

Interpretation. The event shifts demand because index trackers must rebalance when index weights change. It is not a conventional index addition, secondary offering, or information announcement.

2.2 Test and comparison samples

- The **test sample** contains 31 TSE 300 stocks whose public floats and weights increased because of the redefinition.
- The **comparison sample** contains 261 TSE 300 stocks whose public floats were not directly affected.
- Six potential comparison stocks with confounding float increases are excluded.
- The paper also examines a **nonindex comparison sample**: 38 nonindex stocks whose floats changed but which were not in the TSE 300.

Interpretation. The comparison sample controls for market-wide returns. The nonindex sample tests whether the float definition itself conveyed information.

2.3 Ownership, quote, return, and volume data

- Ownership and public-float data come from TSE 300 Composite Index Summary files.
- Bid/ask quotes, transaction prices, and trading volume come from monthly intraday TSE transaction records.
- Insider trading and equity issuance are checked using OSC Monthly Bulletins.
- Confounding news is checked using *The Globe and Mail* CD-ROM index.

Interpretation. The paper verifies that the float changes did not come from insider sales, equity issuance, or firm-specific news.

2.4 Demand-shift variables

The paper uses three measures of the demand shock:

- $\Delta Float_j$: percentage change in stock j 's public float.
- $\Delta Weight_j$: percentage change in stock j 's TSE 300 index weight.
- $DaysTraded_j$: float change divided by mean daily trading volume, multiplied by the share of the TSE 300 held in indexed portfolios.

Interpretation. $\Delta Float$ and $\Delta Weight$ measure the size of the index-induced demand shift. $DaysTraded$ measures how hard it is for normal trading volume to absorb the shock.

2.5 Return and turnover variables

- Week 0 is the week ending Friday, November 15, 1996.
- Excess return for a test stock is its return minus the mean return of the comparison sample.
- Beta-adjusted abnormal returns are also reported for the test sample.
- Turnover is weekly volume divided by shares outstanding.
- Excess turnover is weekly turnover divided by normal turnover, minus one; normal turnover is median turnover over weeks -5 through -2 .

Interpretation. Returns measure the price impact of the demand shock. Excess turnover measures rebalancing pressure and when that pressure dissipates.

3 Models and empirical specifications

3.1 Three competing hypotheses

- **Downward-sloping demand curves:** a non-informational demand increase permanently raises price.
- **Information:** price rises because the event conveys favorable information.
- **Short-term price pressure:** price rises temporarily because liquidity suppliers require compensation during the rebalancing window.

Interpretation. The TSE event is useful because it separates these explanations more cleanly than index additions or secondary equity distributions.

3.2 Event-week excess return

For stock j in week t , define

$$ER_{j,t} = R_{j,t} - \bar{R}_t^{comp}. \quad (1)$$

Week 0 is the event week, November 11–15, 1996.

Interpretation. A positive mean $ER_{j,0}$ for the 31 affected stocks measures the price effect beyond broad TSE 300 market movement.

3.3 Excess turnover

Let $Turnover_{j,t}$ be weekly volume divided by shares outstanding. Then

$$ExcessTurnover_{j,t} = \frac{Turnover_{j,t}}{\text{median}(Turnover_{j,-5}, \dots, Turnover_{j,-2})} - 1. \quad (2)$$

Interpretation. If index funds rebalance, excess turnover should spike around week 0 and then subside. If price pressure is temporary, returns should reverse after this spike ends.

3.4 Reversal regression

To test for price reversal, the paper estimates

$$CAR_{1-T,j} = \alpha + \theta AR_{0,j} + \varepsilon_{t-T,j}, \quad (3)$$

where $CAR_{1-T,j}$ is cumulative abnormal return from week +1 to week + T , and $AR_{0,j}$ is the event-week abnormal return.

- Complete reversal predicts $\alpha = 0$ and $\theta = -1$.
- No reversal predicts $\theta = 0$.

Interpretation. This is a firm-by-firm test of whether stocks with the largest event-week price increases later give back those gains.

3.5 Cross-sectional SUR regressions

The paper estimates the following system:

$$AR_j = b_0 + b_1 \Delta Demand_j + b_2 CapDum_j + b_3 \ln(MV_j) + \varepsilon_j, \quad (4)$$

$$ExcessTurnover_j = c_0 + c_1 |\Delta Demand_j| + c_2 CapDum_j + c_3 \ln(MV_j) + \eta_j. \quad (5)$$

Here $\Delta Demand_j$ is alternatively $\Delta Float_j$, $\Delta Weight_j$, or $DaysTraded_j$; $CapDum_j = 1$ if the stock rose by more than 50% from January 1 to October 31, 1996.

Interpretation. Positive b_1 means larger demand shifts cause larger price changes. Positive c_1 means larger shifts cause more trading. A positive return effect combined with no later reversal supports downward-sloping demand.

4 Identification and empirical design

4.1 Why the event is close to information-free

- It was announced three months before implementation.
- The ownership data needed to infer affected firms were publicly available.
- The 20% threshold matched the OSC definition, so the TSE change had no legal effect on shareholders.
- No new firms entered the index, so index-certification effects are absent.
- The authors find no abnormal returns or volume around the August 6 announcement.
- The nonindex float-change sample does not earn event-week abnormal returns.

Interpretation. This is the paper’s main identification strength: the demand shift is mechanical, predictable, and separated from firm news.

4.2 Why short-term price pressure is unlikely

- Excess turnover spikes around week 0 and normalizes soon after.
- Event-week returns do not reverse when volume returns to normal.
- Reversal regressions reject complete reversal through about week +9.
- The point estimate of cumulative excess returns at week +15 remains close to the event-week effect.

Interpretation. Short-term price pressure predicts that prices should fall back once index funds complete rebalancing. The evidence does not show that pattern.

4.3 What the paper cannot fully prove

- The treated sample contains only 31 stocks.
- The event occurs at a common date, so cross-sectional standard errors may understate dependence.
- Long-horizon reversal tests lose power as the cumulative window grows.
- Very long-lived price pressure cannot be completely ruled out statistically.
- The paper observes trading volume, not the exact trades of each index fund.

Interpretation. The evidence is strongest against information and short-term pressure. It supports downward-sloping demand, while acknowledging limited power at long horizons.

5 Tables and figures

The paper contains no original graphs. The core evidence is in five tables, directly extracted below and grouped where possible to save space.

5.1 Table I + Table II: demand shock and event-week returns

Read: Table I:

- The test sample has 31 affected stocks; the comparison group has 261 stocks.
- Mean public float rises from 46.4 million to 58.2 million shares for the test sample.
- Mean $\Delta Float/Outstanding$ is 19.33%; median is 17.51%.
- Mean TSE 300 weight rises from 0.2539 to 0.3135 times 10^{-2} .
- Median *DaysTraded* is 6.00 for the test sample versus 0.01 for the comparison group.

Read: Table II:

- Week 0 mean return is 4.31% for test stocks and 1.97% for comparison stocks.
- Week 0 mean excess return is 2.34%, with $p = 0.01$.
- Week 0 beta-adjusted abnormal return is 2.07%, also with $p = 0.01$.
- Median test-sample return is 3.87% versus 1.40% for comparison stocks.
- Post-event weekly differences are small and insignificant.

Economic message. The redefinition creates an economically meaningful index-demand shock, and affected stocks rise exactly when index weights increase.

Float is the number of shares in noncontrolling hands. The pre-redefinition float is measured on October 31, 1996 and includes all shares held in stakes under 15 percent. The post-redefinition float is measured on November 15, 1996 and includes shares held in stakes under 20 percent. Days traded is calculated as the ratio of the float change to mean daily trading volume during August through October 1996, multiplied by the fraction of the value of TSE 300 firms held in indexed portfolios. Weight is the TSE 300 weight on October 31 and November 15, 1996. The test group consists of 31 firms whose public floats are affected by the TSE redefinition of controlling blocks. The comparison group consists of 261 firms whose floats are not affected by the redefinition. p-values for changes in variables from pre-redefinition to post-redefinition are in parentheses.				
	Mean	First Quartile	Median	Third Quartile
Panel A: Test Sample ($N = 31$)				
Firm size (market value, \$million)	1,269	298	516	762
Pre-redefinition float	46,423,173	18,087,592	25,195,360	41,433,837
Post-redefinition float	58,213,858	23,557,245	30,485,764	52,047,501
$\Delta Float$	11,790,685	4,223,550	7,983,199	10,207,321
	(0.000)		(0.000)	
$\Delta Float/Outstanding$ shares	0.1933	0.1572	0.1751	0.1939
	(0.000)		(0.000)	
Days traded	6.98	3.09	6.00	9.24
Pre-redefinition weight, $\times 10^{-2}$	0.2539	0.0600	0.1000	0.1800
Post-redefinition weight, $\times 10^{-2}$	0.3135	0.0700	0.1300	0.3000
$\Delta Weight/Pre$ -redefinition weight	0.2872	0.1818	0.2333	0.3333
	(0.000)		(0.000)	
Panel B: Comparison Group ($N = 261$)				
Firm size (market value, \$million)	1,774	348	659	1,668
Pre-redefinition float	59,737,049	18,854,120	37,332,909	73,113,192
Post-redefinition float	60,214,364	19,318,997	37,582,905	77,131,960
$\Delta Float$	477,315	0	9,100	98,437
	(0.000)		(0.000)	
$\Delta Float/Outstanding$ shares	0.0002	0.0000	0.0000	0.0000
	(0.000)		(0.000)	
Days traded	0.34	0	0.01	0.10
Pre-redefinition weight, $\times 10^{-2}$	0.3505	0.0700	0.1400	0.3100
Post-redefinition weight, $\times 10^{-2}$	0.3422	0.0700	0.1400	0.2900
$\Delta Weight/Pre$ -redefinition weight	-0.0291	-0.0714	0.0000	0.0000
	(0.000)		(0.000)	

(a) Table I: float and index-weight changes.

The test group consists of 31 firms whose public floats are directly affected by the TSE redefinition of controlling blocks. The comparison group consists of 261 firms whose floats are not affected by the ruling. p-values from two-sample difference in means tests and two-sample difference in median tests are provided in parentheses. The TSE redefinition became effective Friday, November 15, 1996. The week ending November 15 is designated as week 0.					
Week	Period	Test Group Return	Comparison Group Return	Difference between Test and Comparison Group Return	Beta-Adjusted Abnormal Return for Test Group
Panel A: Means					
Week -1	4/11/96 through 8/11/96	0.0114	0.0174	-0.0060 (0.67)	-0.0062 (.66)
Week 0	11/11/96 through 15/11/96	0.0431	0.0197	0.0234 (0.01)	0.0207 (0.01)
Week +1	18/11/96 through 22/11/96	0.0106	0.0104	0.0002 (0.98)	0.0001 (0.99)
Week +2	25/11/96 through 29/11/96	0.0079	0.0066	0.0013 (0.89)	-0.0004 (0.96)
Week +3	2/12/96 through 6/12/96	-0.0309	-0.0273	-0.0036 (0.75)	-0.0003 (0.98)
Week +4	9/12/96 through 13/12/96	-0.0126	-0.0180	0.0054 (0.39)	0.0047 (0.48)
Week +5	16/12/96 through 20/12/96	0.0224	0.0153	0.0071 (0.44)	0.0042 (0.66)
Week +6	23/12/96 through 27/12/96	0.0219	0.0193	0.0026 (0.81)	-0.0014 (0.90)
Panel B: Medians					
Week -1	4/11/96 through 8/11/96	0.0124	0.0169	-0.0045 (0.67)	-0.0001 (0.77)
Week 0	11/11/96 through 15/11/96	0.0387	0.0140	0.0247 (0.00)	0.0190 (0.02)
Week +1	18/11/96 through 22/11/96	0.0108	0.0034	0.0074 (0.92)	0.0006 (0.74)
Week +2	25/11/96 through 29/11/96	0.0085	0.0018	0.0067 (0.73)	-0.0002 (0.85)
Week +3	2/12/96 through 6/12/96	-0.0252	-0.0268	0.0016 (0.72)	-0.0002 (0.96)
Week +4	9/12/96 through 13/12/96	-0.0074	-0.0140	0.0066 (0.59)	0.0128 (0.26)
Week +5	16/12/96 through 20/12/96	0.0176	0.0091	0.0085 (0.49)	0.0001 (0.85)
Week +6	23/12/96 through 27/12/96	0.0147	0.0138	0.0009 (0.84)	-0.0046 (0.94)

(b) Table II: weekly returns around the event.

5.2 Table III + Table IV: turnover spike and no return reversal

Read: Table III:

- Event-week median excess turnover is 1.000 for test stocks, meaning turnover is twice normal.
- Event-week median excess turnover is 0.395 for comparison stocks.
- The median difference is 0.605 and significant at the 5% level.
- Excess turnover remains elevated in week +1 but becomes normal afterward.

Read: Table IV:

- Complete reversal predicts a slope of -1 in the reversal regression.
- The paper rejects $\theta = -1$ through week +9.
- The coefficient is significantly different from zero only around week +4.
- Long-horizon point estimates remain far from -1 , even when statistical power declines.

Economic message. Trading pressure is temporary, but price effects persist. This combination is inconsistent with a purely short-lived liquidity-pressure story.

Table III
Excess Turnover for TSE 300 Companies Surrounding the Toronto Stock Exchange's Redefinition of the Public Float

Turnover is calculated as the ratio of weekly trading volume to the number of shares outstanding. Excess turnover is the ratio of turnover in any week to the stock's normal weekly turnover (median weekly turnover in week -5 through week -2) less one. The test group consists of 31 firms whose floats are affected by the TSE redefinition of controlling blocks. The comparison group consists of 261 firms whose floats are not affected by the redefinition. The TSE redefinition became effective Friday, November 15, 1996. The week ending November 15 is designated week 0.

Week	Period	Median of Test Group	Median of Comparison Group	Difference between Test and Comparison Sample Medians
Week -1	4/11/96 through 8/11/96	+0.499	+0.376	+0.123
Week 0	11/11/96 through 15/11/96	+1.000	+0.395	+0.605*
Week +1	18/11/96 through 22/11/96	+0.699	+0.396	+0.303
Week +2	25/11/96 through 29/11/96	+0.307	+0.057	+0.250
Week +3	2/12/96 through 6/12/96	+0.110	+0.281	-0.171
Week +4	9/12/96 through 13/12/96	+0.050	+0.149	-0.099
Week +5	16/12/96 through 20/12/96	+0.443	+0.355	+0.088
Week +6	23/12/96 through 27/12/96	-0.287	-0.434	+0.147

* denotes significance at the 5-percent level based on the Wilcoxon two-sample test of difference

(a) Table III: excess turnover.

Table IV
Regression Estimates in a Test of Return Reversals for Test Sample Firms

For varying post-event windows, the following cross-sectional regression is estimated for 31 firms whose public floats are affected by the TSE's redefinition (the test group).

$$CAR_{1-T,j} = \alpha + \theta AR_{0,j} + \epsilon_{1-T,j}.$$

The dependent variable, $CAR_{1-T,j}$, is the cumulative abnormal stock return beginning in week +1 through week $+T$ inclusive. The independent variable, $AR_{0,j}$, is the abnormal return in week 0. The TSE redefinition became effective Friday, November 15, 1996. The week ending November 15 is designated week 0. The coefficient on $AR_{0,j}$ equals -1 under the price pressure prediction of complete reversals. The coefficient on $AR_{0,j}$ equals zero under the hypothesis of no reversals.

Dependent Variable	α	θ	p-value: $\theta = -1$	p-value: $\theta = 0$	R^2
$CAR_{1-1,j}$	-0.0032	0.1621	0.00*	0.47	0.018
$CAR_{1-2,j}$	-0.0042	0.1910	0.00*	0.47	0.018
$CAR_{1-3,j}$	0.0046	-0.2588	0.01*	0.33	0.033
$CAR_{1-4,j}$	0.0148	-0.5394	0.01*	0.03*	0.144
$CAR_{1-5,j}$	0.0167	-0.4373	0.01*	0.13	0.076
$CAR_{1-6,j}$	0.0152	-0.4277	0.05*	0.33	0.033
$CAR_{1-7,j}$	0.0129	-0.2970	0.03*	0.48	0.017
$CAR_{1-8,j}$	0.0015	0.1593	0.05*	0.84	0.002
$CAR_{1-9,j}$	-0.0002	0.1283	0.08	0.87	0.001
$CAR_{1-10,j}$	-0.0010	0.0130	0.15	0.98	0.000
$CAR_{1-11,j}$	0.0007	-0.1008	0.21	0.91	0.001

(b) Table IV: reversal regressions.

5.3 Table V: cross-sectional demand-shift regressions

Read:

- $\Delta Float$ has positive coefficients in abnormal-return regressions: 0.1048 and 0.0955.
- $\Delta Weight$ also has positive abnormal-return coefficients: 0.1114 and 0.1078.
- $DaysTraded$ is positive but insignificant in abnormal-return regressions.
- In the turnover regressions, $|\Delta Float|$, $|\Delta Weight|$, and $|DaysTraded|$ are positive; most are statistically significant.
- $CapDum$ is positive in return regressions and negative in turnover regressions, consistent with low supply elasticity for stocks with large capital gains.
- The coefficient 0.0955 on $\Delta Float$ implies a rough excess-demand elasticity estimate of about 10.5.

Economic message. Larger index-induced demand shifts are associated with larger price increases and more trading. This is the cross-sectional signature of downward-sloping stock demand curves.

Table V
Seemingly Unrelated Regression Estimates

The regressions below are estimated as a system using the seemingly unrelated regressions estimation procedure. The dependent variables in the two models are the week 0 abnormal return and the week 0 excess turnover for 292 stocks in the TSE 300 index. The TSE redefinition became effective Friday, November 15, 1996. The week ending November 15 is designated week 0. ΔFloat and ΔWeight measure the percentage change in floats and index weights. Days traded is the ratio of the float change to mean daily trading volume. Absolute values of ΔFloat , ΔWeight , and Days traded are used in Model 2 (dependent variable: Excess Turnover). CapDum is an indicator variable that equals one if the stock price increases by more than 50 percent during January 1, 1996 through October 31, 1996, and zero otherwise. $\text{Ln}(\text{MV})$ is the natural log of the firm's market value. p -values are given in parentheses. Estimated cross-equation error correlations are significantly different from zero in all regressions.

	Reg. #1	Reg. #2	Reg. #3	Reg. #4	Reg. #5	Reg. #6
Model 1						
Dependent Variable: Abnormal Return						
Intercept	-0.0034 (0.28)	-0.0007 (0.81)	-0.0003 (0.93)	-0.0194 (0.26)	-0.0156 (0.34)	-0.0142 (0.41)
ΔFloat	0.1048* (0.00)			0.0955* (0.00)		
ΔWeight		0.1114* (0.00)			0.1078* (0.00)	
Days traded			0.0005 (0.53)			0.0005 (0.51)
CapDum				0.0136 (0.07)	0.0144* (0.04)	0.0182* (0.02)
$\text{Ln}(\text{MV})$				0.0021 (0.43)	0.0018 (0.46)	0.0015 (0.56)
Model 2						
Dependent Variable: Excess Turnover						
Intercept	0.7592* (0.00)	0.7134* (0.00)	0.7737* (0.00)	1.6661* (0.00)	1.5780* (0.00)	1.6573* (0.00)
$ \Delta\text{Float} $	1.1455 (0.06)			1.2434* (0.04)		
$ \Delta\text{Weight} $		1.0420* (0.02)			0.9942* (0.02)	
$ \text{Days traded} $			0.0388* (0.01)			0.0352* (0.02)
CapDum				-0.3169* (0.03)	-0.2976* (0.03)	-0.2629 (0.06)
$\text{Ln}(\text{MV})$				-0.1281* (0.01)	-0.1210* (0.01)	-0.1254* (0.01)
System weighted R^2	0.022	0.067	0.012	0.055	0.098	0.047

* denotes significance at the 5 percent level.

Figure 3: Table V: SUR regressions of abnormal returns and excess turnover.

6 Short summary

- The paper tests whether individual stock demand curves slope down.
- The setting is the TSE 300 public-float redefinition implemented on November 15, 1996.
- The rule change mechanically increased floats and index weights for 31 stocks.
- Index funds tracking the TSE 300 had to buy more of the affected stocks.
- The event was unlikely to convey information because it was pre-announced, used public data, matched the OSC definition, and did not involve index additions.
- The affected stocks earn an event-week excess return of 2.34%.
- Beta-adjusted abnormal return is 2.07% in week 0.
- Event-week turnover is unusually high, consistent with rebalancing.
- A nonindex float-change sample does not earn comparable abnormal returns.
- Bid-ask spreads do not change materially, arguing against liquidity improvements as the source.
- Turnover returns to normal quickly, but price increases do not reverse quickly.
- Reversal regressions reject complete reversal through week +9.

- Cross-sectional regressions show that larger demand shifts predict larger price changes and more trading.
- The evidence supports downward-sloping demand curves for stocks.

7 Conclusion

7.1 Core conclusion

Kaul, Mehrotra, and Morck show that a mechanical index-weight demand shock raises stock prices even when the event contains little or no information. This is strong evidence that demand curves for individual stocks slope downward.

7.2 Economic intuition

If stocks had perfect substitutes, a rightward shift in demand from index funds would be absorbed at essentially unchanged prices. Instead, prices rise when index weights rise, and the price increase persists after rebalancing volume fades. Investors therefore do not treat the affected stocks as perfectly replaceable claims.

7.3 Takeaway

The paper finds a clean event than standard index additions. The TSE redefinition changes demand without adding firms, certifying firms, or revealing obvious news. The result is a direct challenge to the assumption that stock demand curves are flat.